

Agents

- A - Alice
- B - Browser with Session storage, SubtleCrypto API, and libsodium library
- M - Webauthn authenticator (passkey manager)
- S - Quick Crypt server

Browser and Libsodium Functions

- D_H - BLAKE2b-512 KDF from libsodium
- D_P - PBKDF2-HMAC-SHA512 key derivation FIPS-180-4 from SubtleCrypto
- D_S - HMAC-SHA-512 from SubtleCrypto
- E_a - Symmetric cipher using AEAD algorithm a . One of:
 - 1. AES-256 in Galois/Counter mode from SubtleCrypto
 - 2. XChaCha20 with Poly1305 MAC from libsodium
 - 3. AEGIS 256 from libsodium
- G - HMAC-SHA-512 key generator from SubtleCrypto
- H - BLAKE2b keyed hash (MAC) generator from libsodium
- H_2 - SHA-256 from SubtleCrypto
- P_K - ML-DSA-65 (FIPS 204) keypair generator from libcrux
- P_S - ML-DSA-65 (FIPS 204) randomized signature generator from libcrux
- P_V - ML-DSA-65 (FIPS 204) signature verifier from libcrux
- R - Cryptographic pseudo-random generator from libsodium
- V - Constant-time hash (MAC) validator from libsodium

Cipher Variables

a - Symmetric AEAD cipher and mode: [1, 2, 3]
 ad_F - Additional data included in ciphertext
 ad_X - ad_F with extended context
 b - Valid or invalid MAC tag
 cd - Cipher data
 cd_0 - Cipher data block 0
 cd_N - Cipher data block N
 cd_u - u_c encrypted under session key k_L
 err - Error message and exit
 f - Block flags
 h - Password hint text
 h_E - Encrypted hint
 h_El - Encrypted hint length
 i - PBKDF2-HMAC-SHA512 iteration count: min 420,000 max 4,294,000,000
 k_H - 256 bit ephemeral hint cipher key
 k_{M0} - 256 bit ephemeral message block0 cipher key
 k_{MN} - 256 bit ephemeral message blockN cipher key
 k_S - 256 bit ephemeral MAC key
 k_C - 256 bit ephemeral commit key
 kp_B - Key purpose text: "Blck_Key"
 kp_C - Key purpose text: "Cmit_Key"
 kp_H - Key purpose text: "Hint_Key"
 kp_S - Key purpose text: "Sign_Key"
 l - Payload length
 le - Loop end (1-16)
 lp - Loop count (1-16)
 m - Clear text
 m_0 - Clear text block 0
 m_N - Clear text block N
 m_E - Block of encrypted message
 n_{IV} - Pseudo random initialization vector
 $n_{IV}l$ - n_{IV} bit length: [96, 192, 256]
 n_S - 128 bit pseudo random salt
 N - Block number
 p - Password text
 r - 384 bits of pseudo random data
 t - 256 bit MAC tag
 t_L - Last 256 bit MAC tag
 u_c - 256 bit user credential in memory
 v - Cipher data version: 7

Message Encryption by A

$A \xleftrightarrow{\text{webauthn}} B, M \xleftrightarrow{\text{webauthn}} S$
 $B : u_c = cd_u$ decryption under k_L
 $A \rightarrow B : m, i, le$
 $v = 7$
 $lp = 1$
 $t_L = 0x00$
LOOP : B compute
 $A \rightarrow B : p, h, a$
 $r = R(384)$
 $n_S = r[0 : 128]$
 $n_{IV} = r[128 : 128 + n_{IV}l]$
 $n_{IVH} = D_H(2, kp_H, H(n_S, a, v, lp, n_{IV}))$
 $k_H = D_H(1, kp_H, H(n_S, a, v, lp, u_c))$
 $k_S = D_H(1, kp_S, H(n_S, a, v, lp, u_c))$
 $k_{M0} = D_P(p \parallel u_c \parallel a \parallel v \parallel lp, n_S, i)$
 $k_C = D_H(1, kp_C, H(n_S, k_{M0}))$
 $m_0 \parallel \dots \parallel m_N = m$
 $cd = cd_0 \parallel \dots \parallel cd_N$
 $m = cd$
 $lp = lp + 1$
 goto *LOOP* if $lp \leq le$
 $A \leftarrow B : cd$

cd_0 Encryption by B

$h_E = E_a(h, n_{IVH}, \emptyset, k_H)$
 $h_El = len(h_E)$
 $f = 1$ if TERM else 0
 $ad_F = f \parallel a \parallel n_{IV} \parallel n_S \parallel i \parallel le \parallel lp \parallel h_El \parallel h_E$
 $ad_X = ad_F \parallel k_C$
 $m_E = E_a(m_0, n_{IV}, ad_X, k_{M0})$
 $l = len(ad_F \parallel m_E)$
 $t = H(v \parallel l \parallel ad_F \parallel m_E \parallel t_L, k_S)$
 $t_L = t$
 $cd_0 = t \parallel v \parallel l \parallel ad_F \parallel m_E$

cd_N Encryption by B

$r = R(384)$
 $n_{IV} = r[0 : n_{IV}l]$
 $f = 1$ if TERM else 0
 $ad_F = f \parallel a \parallel n_{IV}$
 $k_{MN} = D_H(N, kp_B, H(n_S, k_{M0}))$
 $ad_X = ad_F \parallel k_C$
 $m_E = E_a(m_N, n_{IV}, ad_X, k_{MN})$
 $l = len(ad_F \parallel m_E)$
 $t = H(v \parallel l \parallel ad_F \parallel m_E \parallel t_L, k_S)$
 $t_L = t$
 $cd_N = t \parallel v \parallel l \parallel ad_F \parallel m_E$

Message Decryption by A

$A \xleftrightarrow{\text{webauthn}} B, M \xleftrightarrow{\text{webauthn}} S$
 $B : u_c = cd_u$ decryption under k_L
 $A \rightarrow B : cd$
 $t_L = 0x00$
 $LOOP : B$ compute
 $cd_0 \parallel \dots \parallel cd_N = cd$
 $t, v, l, ad_F, m_E = cd_0$
 $f, a, n_{IV}, n_S, i, le, lp, h_E l, h_E = ad_F$
 $k_S = D_H(1, kp_S, H(n_S, a, v, lp, u_c))$
 $b = V(v \parallel l \parallel ad_F \parallel m_E \parallel t_L, k_S, t)$
 $t_L = t$
 if !b :
 $A \leftarrow B : err$
 $n_{IVH} = D_H(2, kp_H, H(n_S, a, v, lp, n_{IV}))$
 $k_H = D_H(1, kp_H, H(n_S, a, v, lp, u_c))$
 $h = E_a^{-1}(h_E, n_{IVH}, \emptyset, k_H)$
 $B \rightarrow A : h$
 $B \leftarrow A : p$
 $k_{M0} = D_P(p \parallel u_c \parallel a \parallel v \parallel lp, n_S, i)$
 $k_C = D_H(1, kp_C, H(n_S, k_{M0}))$
 $m = m_0 \parallel \dots \parallel m_N$
 $cd = m$
 goto LOOP if lp > 1
 $A \leftarrow B : m$

m_0 Decryption by B

$ad_X = ad_F \parallel k_C$
 $m_0 = E_a^{-1}(m_E, n_{IV}, ad_X, k_{M0})$

m_N Decryption by B

$t, v, l, ad_F, m_E = cd_N$
 $f, a, n_{IV} = ad_F$
 $b = V(v \parallel l \parallel ad_F \parallel m_E \parallel t_L, k_S, t)$
 $t_L = t$
if !b :
 $A \leftarrow B : err$
 $k_{MN} = D_H(N, kp_B, H(n_S, k_{M0}))$
 $ad_X = ad_F \parallel k_C$
 $m_N = E_a^{-1}(m_E, n_{IV}, ad_X, k_{MN})$

u_c Encryption at Rest Variables

a - Symmetric AEAD cipher and mode: 2
 ad_F - Additional data included in ciphertext
 ad_X - ad_F with extended context
 b - Valid or invalid MAC tag
 cd_p - u_c encrypted under PRF output p_R
 cd_r - u_c encrypted under recovery secret $u_r \parallel u_i$
 cd_u - u_c encrypted under session key k_L
 cd_x - Encrypted u_c cipher data: one of cd_u, cd_p, cd_r
 err - Error message and exit
 f - Block flags: 1 (fixed)
 h_{El} - Encrypted hint length: 0 (unused)
 i - PBKDF2 iteration count: 0 (unused)
 k - 256 bit master key: one of k_L, p_R , or $u_r \parallel u_i$
 k_C - 256 bit ephemeral commit key
 k_{KS} - HMAC-SHA-512 non-extractable CryptoKey, stored in IndexedDB
 k_L - 256 bit ephemeral per session master key
 k_M - 256 bit ephemeral message cipher key
 k_S - 256 bit ephemeral MAC key
 kp_C - Key purpose text: "Cmit_Key"
 kp_M - Key purpose text: "Cphr_Key"
 kp_S - Key purpose text: "Sign_Key"
 l - Payload length
 le - Loop end: 1 (fixed)
 lp - Loop count: 1 (fixed)
 m_E - Encrypted message
 n_{IV} - Pseudo random initialization vector
 $n_{IV}l$ - n_{IV} bit length: 192
 n_S - 128 bit pseudo random salt
 p_R - 256 bit passkey PRF output
 $pkId$ - Passkey credential identifier
 r - 320 bits of pseudo random data
 s - IndexedDB slot label: "user-cred-key"
 t - 256 bit MAC tag
 u_c - 256 bit user credential (plaintext)
 u_i - User id
 u_r - 128 bit recovery id
 v - Cipher data version: 7

k_L Derivation by B

$B \leftarrow M : pkId$
 $k_{KS} = G()$
 $k_L = D_S(s \parallel pkId, k_{KS})[32 : 64]$

u_c Encryption Under k by B

where $(k, cd_x) \in \{(k_L, cd_u), (p_R, cd_p), (u_r \parallel u_i, cd_r)\}$
 $r = R(320)$
 $n_S = r[0 : 128)$
 $n_{IV} = r[128 : 128 + n_{IV}l)$
 $k_M = D_H(0, kp_M, H(n_S, a, v, lp, u_i, k))$
 $k_S = D_H(1, kp_S, H(n_S, a, v, lp, u_i, k))$
 $k_C = D_H(1, kp_C, H(n_S, k_M))$
 $ad_F = f \parallel a \parallel n_{IV} \parallel n_S \parallel i \parallel le \parallel lp \parallel h_E l$
 $ad_X = ad_F \parallel u_i \parallel k_C$
 $m_E = E_a(u_c, n_{IV}, ad_X, k_M)$
 $l = len(ad_F \parallel m_E)$
 $t = H(v \parallel l \parallel ad_F \parallel m_E \parallel 0x00, k_S)$
 $cd_x = t \parallel v \parallel l \parallel ad_F \parallel m_E$

cd_x Decryption Under k by B

where $(k, cd_x) \in \{(k_L, cd_u), (p_R, cd_p), (u_r \parallel u_i, cd_r)\}$
 $B \leftarrow S : u_i$
 $t, v, l, ad_F, m_E = cd_x$
 $f, a, n_{IV}, n_S, i, le, lp, h_E l = ad_F$
 $k_S = D_H(1, kp_S, H(n_S, a, v, lp, u_i, k))$
 $b = V(v \parallel l \parallel ad_F \parallel m_E \parallel 0x00, k_S, t)$
if ! b :
 $A \leftarrow B : err$
 $k_M = D_H(0, kp_M, H(n_S, a, v, lp, u_i, k))$
 $k_C = D_H(1, kp_C, H(n_S, k_M))$
 $ad_X = ad_F \parallel u_i \parallel k_C$
 $u_c = E_a^{-1}(m_E, n_{IV}, ad_X, k_M)$

Authentication Variables

ao - Authentication options, including *o*, *ch*
ar - Authentication response, including signed *ch*
bd - HTTP request body
cd_p - *u_c* encrypted under PRF output *p_R*
cd_r - *u_c* encrypted under recovery secret *u_r* || *u_i*
cd_u - *u_c* encrypted under session key *k_L*
ch - 256 bit challenge value
ch_R - 256 bit recovery challenge value
ck - Session cookie, a signed JWT
cs - CSRF token
cw - Alice's webauthn authenticator credentials
kp_R - Key purpose text: "RecovKey"
kp_U - Key purpose text: "UCredKey"
m_R - Recovery proof message
m_U - userCred proof message
meth - HTTP request method
n_P - 256 bit proof nonce value
o - Quick Crypt origin "https://quickcrypt.org"
p_R - 256 bit passkey PRF output
p_S - Fixed PRF salt
pk_R - ML-DSA-65 recovery proof public key
pk_U - ML-DSA-65 userCred proof public key
pth - HTTP request path
ro - Registration options, including *o*, *ch*, *u_i*
rq - Authorized request, including *meth*, *pth*, *bd*
rr - Registration response, including signed *ch*
sc_R - Recovery signature context: "qcrypt/recovery/proof/v1"
sc_U - userCred signature context: "qcrypt/usercred/proof/v1"
sk_R - ML-DSA-65 recovery proof secret key
sk_U - ML-DSA-65 userCred proof secret key
ts_R - Recovery proof timestamp in milliseconds
ts_U - userCred proof timestamp in milliseconds
u_c - 256 bit user credential in memory
u_i - 128 bit user id guaranteed to be unique
u_n - A's chosen user name
u_r - 128 bit recovery id
 Δ - Maximum request timestamp skew: 120 seconds
 δ - Delimiter byte: \n
 σ_R - ML-DSA-65 recovery proof signature
 σ_U - ML-DSA-65 userCred proof signature

Registration by A

$A \rightarrow B : u_n$
 $B \rightarrow S : u_n, o$
 $S : u_i = R(128); ch = R(256); \text{ store } u_i, ch$
 $B \leftarrow S : ro$
 $B \rightarrow M : ro, p_S$
 $A \rightarrow M : cw$
 M create and store passkey, sign ch
 $B \leftarrow M : rr, p_R$
 $B : u_r = R(128)$
 $B : (pk_R,) = P_K(D_H(1, kp_R, u_r \parallel u_i))$
 $B : u_c = R(256)$
 $B : cd_p = u_c$ encryption under p_R
 $B : cd_r = u_c$ encryption under $u_r \parallel u_i$
 $B : (pk_U,) = P_K(D_H(1, kp_U, u_c))$
 $B \rightarrow S : rr, u_i, ch, pk_R, pk_U, cd_p, cd_r$
 S verify webauthn signature; store $rr, pk_R, pk_U, cd_p, cd_r$; remove ch
 S create ck, cs
 $B \leftarrow S : u_i, u_n, cs, ck$
 $B : cd_u = u_c$ encryption under k_L ; store cd_u
 $A \leftarrow B : u_r \parallel u_i$ as BIP39

No-PRF fallback: If M returns no p_R , B sends rr, u_i, ch, pk_R only; S generates $u_c = R(256)$, stores it encrypted, derives $(pk_U,)$ from u_c , and returns u_c ; B creates and stores cd_u from u_c .

Authentication by A

$B \rightarrow S : o[, u_i]$
 $S : ch = R(256); \text{ store } ch$
 $B \leftarrow S : ao$
 $B \rightarrow M : ao, p_S$
 $A \rightarrow M : cw$
 M sign ch
 $B \leftarrow M : ar, p_R$
 $B \rightarrow S : ar, ch$
 S verify webauthn signature; remove ch
 S create ck, cs
 $B \leftarrow S : u_i, u_n, cd_p, cs, ck$
 $B : u_c = cd_p$ decryption under p_R
 $B : cd_u = u_c$ encryption under k_L ; store cd_u

No-PRF fallback: S returns u_c in place of cd_p ; B creates and stores cd_u from u_c .

Authorization by B

$B : u_c = cd_u$ decryption under k_L
 $B : (pk_U, sk_U) = P_K(D_H(1, kp_U, u_c))$
 $B : ts_U = \text{current time}$
 $B : n_P = R(256)$
 $B : m_U = u_i\delta \parallel mth\delta \parallel pth\delta \parallel ts_U\delta \parallel n_P\delta \parallel H_2(bd)$
 $B : \sigma_U = P_S(sk_U, m_U, sc_U)$
 $B \rightarrow S : rq, ck, cs, ts_U, n_P, \sigma_U$
 S verify ck, cs
 S verify ts_U within Δ
 $S : m_U = u_i\delta \parallel mth\delta \parallel pth\delta \parallel ts_U\delta \parallel n_P\delta \parallel H_2(bd)$
 $S : b = P_V(pk_U, m_U, \sigma_U, sc_U)$
if !b :
 $B \leftarrow S : err$
 $A \leftarrow B : err$
 $B \leftarrow S : \text{rq response}$

For state changing requests: S additionally requires n_P to be unknown and stores it, rejecting replays.

Add Passkey by A

A is authenticated; requests authorized per Authorization of A
 $B \rightarrow S : u_i, o$
 $S : ch = R(256)$; store ch
 $B \leftarrow S : ro$
 $B : u_c = cd_u$ decryption under k_L
 $B \rightarrow M : ro, p_S$
 $A \rightarrow M : cw$
 M create and store passkey, sign ch
 $B \leftarrow M : rr, p_R$
 $B : cd_p = u_c$ encryption under p_R
 $B \rightarrow S : rr, ch, cd_p$
 S verify webauthn signature; store rr, cd_p ; remove ch

No-PRF fallback: B omits p_S and does not decrypt u_c or send cd_p ; S stores the new passkey without encrypted u_c .

Replace Recovery Words by A

A is authenticated; requests authorized per Authorization of A

B repeat Authentication of A

B : $u_r = R(128)$

B : $(pk_R, sk_R) = P_K(D_H(1, kp_R, u_r \parallel u_i))$

B : ts_R = current time

B : $n_P = R(256)$

B : $m_R = u_i \delta \parallel ts_R \delta \parallel n_P$

B : $\sigma_R = P_S(sk_R, m_R, sc_R)$

B : $u_c = cd_u$ decryption under k_L

B : $cd_r = u_c$ encryption under $u_r \parallel u_i$

B $\rightarrow$ S : $pk_R, ts_R, n_P, \sigma_R, cd_r$

S verify ts_R within Δ

S : $m_R = u_i \delta \parallel ts_R \delta \parallel n_P$

S : $b = P_V(pk_R, m_R, \sigma_R, sc_R)$

if !b :

B $\leftarrow$ S : err

A $\leftarrow$ B : err

S verify n_P unknown; store n_P

S store pk_R, cd_r

B $\rightarrow$ A : $u_r \parallel u_i$ as BIP39

No-PRF fallback: B does not decrypt cd_u or send cd_r ; S stores pk_R without cd_r .

$A \rightarrow B : u_r \parallel u_i$ as BIP39
 $B : (pk_R, sk_R) = P_K(D_H(1, kp_R, u_r \parallel u_i))$
 $B : ts_R = \text{current time}$
 $B : n_P = R(256)$
 $B : m_R = u_i \delta \parallel ts_R \delta \parallel n_P$
 $B : \sigma_R = P_S(sk_R, m_R, sc_R)$
 $B \rightarrow S : u_i, ts_R, n_P, \sigma_R$
 S verify ts_R within Δ
 $S : m_R = u_i \delta \parallel ts_R \delta \parallel n_P$
 $S : b = P_V(pk_R, m_R, \sigma_R, sc_R)$
 if !b :
 $B \leftarrow S : err$
 $A \leftarrow B : err$
 S verify n_P unknown; store n_P
 $S : ch_R = R(256)$; store ch_R
 $B \leftarrow S : ch_R, cd_r$
 $B : u_c = cd_r$ decryption under $u_r \parallel u_i$
 $B : (pk_U, sk_U) = P_K(D_H(1, kp_U, u_c))$
 $B : ts_U = \text{current time}$
 $B : m_U = u_i \delta \parallel mth \delta \parallel pth \delta \parallel ts_U \delta \parallel ch_R \delta \parallel H_2(\emptyset)$
 $B : \sigma_U = P_S(sk_U, m_U, sc_U)$
 $B \rightarrow S : u_i, o, ch_R, ts_U, \sigma_U$
 S verify ts_U within Δ ; remove ch_R
 $S : m_U = u_i \delta \parallel mth \delta \parallel pth \delta \parallel ts_U \delta \parallel ch_R \delta \parallel H_2(\emptyset)$
 $S : b = P_V(pk_U, m_U, \sigma_U, sc_U)$
 if !b :
 $B \leftarrow S : err$
 $A \leftarrow B : err$
 S delete existing rr
 $S : ch = R(256)$; store ch
 $B \leftarrow S : ro$
 $B \rightarrow M : ro, p_S$
 $A \rightarrow M : cw$
 M create and store passkey, sign ch
 $B \leftarrow M : rr, p_R$
 $B : cd_p = u_c$ encryption under p_R
 $B \rightarrow S : rr, u_i, ch, cd_p$
 S verify webauthn signature; store rr, cd_p ; remove ch
 S create ck, cs
 $B \leftarrow S : u_i, u_n, cs, ck$
 $B : cd_u = u_c$ encryption under k_L ; store cd_u

No-PRF fallback: S returns u_c in place of cd_r ; B omits p_S and the cd_r decryption.
 After the new passkey, B sends rr, u_i, ch only and S stores it without encrypted u_c .